

Summarized chain of thought (Spontaneous magnetization in the quantum Heisenberg ferromagnet)

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Original prompt (excerpts)

Problem

For every integer $d \geq 3$ and spin S in $\{1/2, 1, 3/2, \dots\}$, consider the nearest-neighbor isotropic quantum Heisenberg ferromagnet on $\mathbb{Z}^d$ at coupling $J=1$ and zero external field. At a site use $C^{(2S+1)}$ with orthonormal basis e_m , $m = -S, -S+1, \dots, S$, and define $S^z e_m = m e_m$, $S^+ e_m = \sqrt{S(S+1) - m(m+1)} e_{m+1}$, with $S^+ e_S = 0$. Put $S^- = (S^+)^*$, $S^x = (S^+ + S^-)/2$ and $S^y = (S^+ - S^-)/(2i)$. The operator S_x^a acts on site x .

For a finite subset Λ of $\mathbb{Z}^d$, the free-boundary Hamiltonian is

$$H_\Lambda = -\sum_{\{x,y\} \subset \Lambda: |x-y|=1} \sum_{a=x,y,z} S_x^a S_y^a,$$

where each unordered edge is counted once. Let A be the norm completion of the union of all finite-site matrix algebras, and let τ_t be the infinite-volume time evolution obtained as the norm limit on local observables of $\exp(it H_\Lambda) A \exp(-it H_\Lambda)$ along increasing boxes. This finite-range interaction defines the limit independently of the exhausting boxes.

A state ω is a positive linear functional on A with $\omega(1)=1$. For finite $\beta > 0$, call it an equilibrium Gibbs state if it satisfies the β -KMS condition for τ : for all A, B in A there is a bounded function $F(A, B)$, continuous on $0 \leq \text{Im } z \leq \beta$ and analytic in its interior, with $F(A, B)(t) = \omega(A \tau_t(B))$ and $F(A, B)(t+i\beta) = \omega(\tau_t(B)A)$ for real t . Translation invariance means $\omega(T_x(A)) = \omega(A)$ for every lattice translation T_x .

Prove or refute that for every d and S specified above there is a finite $\beta_0(d, S) > 0$ such that for every finite $\beta \geq \beta_0$ there exists a translation-invariant β -KMS state ω with

$$(\omega(S_0^x), \omega(S_0^y), \omega(S_0^z)) \neq (0, 0, 0).$$

The state may depend on β, d, S . The target is a zero-field state with spontaneous magnetization at every sufficiently low positive temperature. It imposes no magnetization requirement on the rotation-invariant mixture and asserts no pointwise-correlation equivalence. A negative resolution must give some fixed d, S and arbitrarily large finite β for which every translation-invariant equilibrium state has zero magnetization.

1 From comparison attempts to a simultaneous pin test

The reasoning begins with long interchange cycles, correlations, field derivatives, and magnetized events of subextensive free-energy cost. Seiringer's account and the spin-wave free-energy result of Correggi–Giuliani–Seiringer sharpen the difficulty: free-energy control alone does not supply the required order parameter [7, 4]. A midpoint-affinity comparison suggests averaging many normalized modifications instead of paying for one long path. The unpredictable paths of Benjamini–Pemantle–Peres provide a related geometric idea [2]. Neither comparison directly supplies a theorem implying Heisenberg order.

The assistant reconstructs Tóth's interchange representation [8]. It also recalls Nishimori-type disorder comparisons, tentatively compares Elboim–Sly's high-dimensional unweighted interchange result with the loop-weighted model here, and considers the generalized-loop frameworks of Aizenman–Nachtergaele and Ueltschi [5, 1, 9]. Complex rotations and the classical octopus inequality likewise fail to close its lower-bound argument [6, 3]. Planting paths incurs exponential color costs; current shifts meet bounded spin capacity; small mean defect density permits trapping walls. An average connectivity identity proves insufficient:

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p relation yields self-consistency with arbitrary phase fraction! Can't bootstrap away because domain walls cost etc.

Deleting labels leads to a conditional one-particle kernel on the remaining slots, called holes. Rank-one transpositions suggest an exterior-power identity and determinant cancellation. Yet signed minors do not directly control the positive event that many test sites lie on internal cycles.

The proposed change is symbolic: start with

$$F(X, Y) = \mathbb{E} \prod_i (X_i + Y_{\pi(i)}).$$

Identifying an eliminated vertex's two variables and extracting their linear coefficient splices the permutation, contributing two when a cycle closes. The proposed contraction preserves stability and hence gives negative dependence for the induced output set. This leads to sparse simultaneous pin tests and a downward induction over pin sets. Short-time leakage may control the required escape capacity, but the geometry and uniform estimates remain to be supplied.

2 A proposed proof through geometry and heat leakage

Spin S is represented by $\ell = 2S$ spin-one-half slots with an on-site symmetric projection. The identity $\mathbf{s}_i \cdot \mathbf{s}_j = P_{ij}/2 - I/4$ gives rate-1/2 exchanges; uniform within-site permutations implement the projection. Under the pin law $\mathbb{P}_D$, cycles missing D have weight two, and $\mathcal{B}_D$ is their union. Boundary slots at every time form D_0 .

For old pins E and a disjoint same-time test set T , put $P = E \cup T$, $m = |T|$. The claimed stability argument gives

$$\mathbb{P}_E(T \subseteq \mathcal{B}_E) \leq (2r)^m,$$

where r is the mean fraction of T whose next pin is again in T , under $\mathbb{P}_P$. Exposing the cycles missing P leaves a hole process subject to every remaining cycle hitting P . The exterior calculation identifies its next-pin first moment.

The proposed induction seeks $\mathbb{P}_E(T \subseteq \mathcal{B}_E) \leq p^m$ for tests with growth at most $L(1+h)^\alpha$. It takes $\alpha = 1/100$, $\nu = 1/(100d)$, $s = p^{-11/10}$, $R \asymp s$, and $\beta \geq 5s$. Mark insertion bounds the n th factorial measure by 2^n times its base value. Sparse branching witnesses then control large blocked components. At regular times, local detours yield a Nash inequality; thickening and filling distant components permits routing flows around their shells.

For the boundary-killed period kernel U , the hitting function satisfies $f = 1$ on T and $U^*f = f$ off T , giving

$$m(1-r) \geq \mathbb{E}_P\langle f, (I-U)f \rangle.$$

Writing $f = 1_T + g$, the assistant bounds both early forward and late reverse leakage. Nash smoothing controls the propagated test mass; a triangular time average absorbs its overlap with g into gradient

dissipation. The residual terms include $R^\alpha s^{-d/2}$, $R^{d+\alpha} p^{2d}$, and R^α/s , all claimed to be $o(p)$. This closes the proposed pin induction.

A boundary-up color condition then has surface-order cost, supplies a magnetized spectral tail, and permits a small-field selection followed by a translation-averaged KMS limit.

The next step is to examine the construction more closely, beginning with conditioning and the signs in the exterior-power identity.

3 Separating exposure from conditional first moments

The next stage concentrates on whether exposing all cycles missing the pins leaves the correct law on the remaining slots. For a deterministic initial label set A_0 , let F_{A_0} require its paths to close and avoid P , and let G_{A_0} require every complementary cycle to hit P . Their conjunction identifies the exposed set uniquely. The weighted measure can therefore be expanded using

$$\sum_{A_0} 1_{F_{A_0}} 2^{\#\text{cycles}(A_0)} 1_{G_{A_0}}.$$

Two clock families separate exchanges involving specified labels from exchanges between holes. The former determine the exposed trajectories; after those trajectories are fixed, the latter operate on deterministic available intervals. Revealing the specified mappings at the end of the period leaves uniform completions. The remaining-cycle condition G_{A_0} still has to be imposed.

Before that condition, the layered permutation Π and its mean K satisfy the proposed identity $\mathbb{E} \bigwedge^k \Pi = \bigwedge^k K$. In a determinant expansion with arbitrary matrix W , cycles confined to nonpin vertices Q give zero. A Schur complement and the terms linear in W identify the conditional mean next-pin kernel as

$$K_{P'P'} + K_{P'Q}(I - K_{QQ})^{-1}K_{QP'}.$$

Here P' retains the test pins and uses finitely many boundary cuts. Positive conditioning probability supplies invertibility; a closed nonpin class would contradict it, so the finite nonpin block is transient. Increasing the boundary cuts densely then passes to continuous boundary pinning, with the exposed data fixed. This identifies a conditional first moment, without identifying the full path law.

The comparison walk refreshes its complementary randomness on each circuit; it does not repeat the actual conditioned loop trajectory.

A second issue is the difference between mass and pointwise control. For the killed transition operator C_u , column sums bound mass, but row sums are also needed for $0 \leq C_u 1_T \leq 1$. Describing killed transitions as averages of partial bijections supplies both. Irregular anchors are paid for individually in expectation, avoiding a union over the entire test set. The flow estimate is applied at the same time as the energy it absorbs, without requiring all-time regularity.

The geometric check also identifies a detail to make explicit: representatives at a surviving site must agree across routed edges if their divergences are to cancel. The state construction is checked on both KMS strip boundaries.

4 Making the routing and stopping rules precise

The investigation tests whether the construction would inadvertently prove more than intended. Eliminating a cycle produces weight two; dividing that bias away need not preserve stability. Negative dependence of pin-output sets is also distinct from positive equilibrium spin correlations. A constant function in a reflecting box and transports that reconcentrate mass provide further tests. Boundary killing and norm loss during diffusion are essential to the proposed responses.

The stopping rule is clarified next. Ignoring interior pins means relaxing the stopping criterion of the same walk, while keeping the exposed environment and its law fixed. Only the complementary randomness is refreshed. The expectation in the quadratic-form bound averages over exposed environments, and all auxiliary constants are fixed before choosing the small parameter p .

For the flow, one surviving slot is chosen at every site used by the routings. A short local connection transfers the requested source to its representative. Finite preimages alone, however, do not justify exchanging sums in an infinite flow. Localization must first show that only $C(1+h)^{d\nu}$ original edges can use a target edge at distance h . This establishes local finiteness before the divergence calculation and retains the combined energy bound CmR^α .

The escape estimate receives the same careful separation of random objects. An actual $\mathbb{P}_P$ picture supplies the forced jumps; fresh independent clocks supply hole motion. Leaving the R -ball requires R chronological spatial marks from these two families. An interval shorter than one period never reuses an actual mark, even across the time cut. Counting ordered choices gives $C^R(2s)^R/R!$, hence exponential decay for $R = C_*s$. Reverse propagation uses adjoints of killed path matrices, not inverse diffusion.

The magnetization argument makes the conditional probability above $S/2$ explicit before applying the surface-cost comparison.

The next checks return to the distinction between the independent completion law and its cycle conditioning, then to hull geometry and dissipation. The heat calculation remains under scrutiny.

5 Testing the construction against traps and correlations

The proposed negative dependence is tested against ferromagnetic correlations. The answer distinguishes the open incoming/output experiment from the closed spin-color equilibrium distribution. The existing restriction is reiterated:

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| In particular we do not assume negative dependence under the path exposure.

Conditioning is reconstructed separately for each deterministic A_0 branch of finite-jump trajectories. The determinant calculation is tied directly to $\det(I - B) = \sum_k (-1)^k \text{tr}(\bigwedge^k B)$, with independent interval blocks and fixed layer relabelings. Dense-cut passage keeps the exposure fixed, and killing includes starts at boundary sites. These details are intended to prevent the conditional first moment from being used as an equality of path laws.

Time-dependent traps raise another question: could forced transports continually return diffused mass to the source? Permutations cannot restore squared norm already lost to diffusion. For a broad harmonic plateau, the same-time flow estimate charges persistent overlap to gradient energy. Both end portions remain necessary because U need not be self-adjoint.

The routing check permits a hull representative to lie in another hull. The map is applied once, rather than repeatedly until reaching a fixed point; divergence is computed from the original preimages. Common canonical slots and a local source correction preserve cancellation. The overlap estimate splits into contributions near each center and a convergent dyadic tail, using $d - 2 - 2d\nu > \alpha$.

Finally, let $\mathcal{D}_u(g_u)$ be half the sum of squared edge gradients on the surviving-slot graph at time u . Exchanging the order of integration produces the positive term $2 \int_s^{2s} W(u) \mathcal{D}_u(g_u) du$. The triangular weights satisfy

$$w(u) = \frac{2(2s - u)}{s^2}, \quad W(u) = \frac{(2s - u)^2}{s^2}, \quad \int_s^{2s} \frac{w(u)^2}{W(u)} du = \frac{4}{s}.$$

Young's inequality gives the required CmR^α/s cost without a temporally regular choice of flows. After rechecking the spectral tail and KMS passage, the assistant turns again to global objections that might escape a local calculation.

6 Closing the estimates and selecting the state

The final stage makes the dependence of the conditional law explicit. The marginal distribution of an exposed history is weighted by the unconditioned probability of G_{A_0} . Given that history, the remaining picture is the independent completion law conditioned on G_{A_0} ; the exterior-power calculation comes before this condition. Pin times are taken modulo β , and positive-duration site visits justify detection by dense boundary cuts.

The sparse-witness count is checked at every radius. Bounded copying multiplicity and bounded congestion control the local Nash comparison. For the global flow, a named-endpoint construction handles an edge joining two hulls, and the $\binom{n+d-1}{d-1}$ nonnegative compositions at level n explain the decay of the outward lattice flow. Only the propagation quantities being averaged need the stated measurability.

The induction error is collected into a single uniform bound,

$$\varepsilon(p) = C[e^{-cs} + R^\alpha(s^{-d/2} + \delta + s^{-1})] = o(p), \quad r \leq \varepsilon(p),$$

where $\delta = CR^d p^{2d} + Ce^{-cR^\xi}$, for fixed $\xi > 0$, bounds the probability of irregularity for one anchor at a fixed time. Choosing p once with $2\varepsilon(p) \leq p$ is intended to close the induction and set $\beta_0 = 5s$. The stability argument also separates the Poisson approximation from the end permutations, whose convergence to uniform is explained through a positive common uniform part.

For $M = \sum_{x \in \Lambda} S_x^z$, the zero-field spectral law $\mathbb{P}_0$ satisfies the claimed surface-cost bound

$$\mathbb{P}_0(M/|\Lambda| \geq S/2) \geq \frac{1}{2} \exp(-C_{\beta,d,S} |\partial\Lambda|).$$

Tilting this law by $e^{\beta h M}$ suppresses magnetization below $S/4$ at fixed $h > 0$ as the volume grows. Growing boxes and fields $h \downarrow 0$ are selected diagonally, then the states are averaged over translations. On each fixed KMS strip, local real-time convergence supplies both boundary values; Gaussian damping and the maximum principle give the analytic limit, extended by norm approximation.

The final response claims a translation-invariant zero-field KMS state with positive magnetization at each sufficiently low positive temperature; $S/4$ appears in the field-selection estimate.

References

- [1] Michael Aizenman and Bruno Nachtergaele. *Geometric Aspects of Quantum Spin States*. Communications in Mathematical Physics **164** (1994), 17–63. <https://doi.org/10.1007/BF02108805>.
- [2] Itai Benjamini, Robin Pemantle, and Yuval Peres. *Unpredictable Paths and Percolation*. The Annals of Probability **26**(3) (1998), 1198–1211. <https://doi.org/10.1214/aop/1022855749>.
- [3] Pietro Caputo, Thomas M. Liggett, and Thomas Richthammer. *Proof of Aldous' spectral gap conjecture*. Journal of the American Mathematical Society **23**(3) (2010), 831–851. <https://doi.org/10.1090/S0894-0347-10-00659-4>.
- [4] Michele Correggi, Alessandro Giuliani, and Robert Seiringer. *Validity of the Spin-Wave Approximation for the Free Energy of the Heisenberg Ferromagnet*. Communications in Mathematical Physics **339**(1) (2015), 279–307. <https://doi.org/10.1007/s00220-015-2402-0>.

- [5] Dor Elboim and Allan Sly. *Infinite cycles in the interchange process in five dimensions*. Preprint, arXiv:2211.17023 (2022; revised 2024). <https://arxiv.org/abs/2211.17023>.
- [6] Oliver A. McBryan and Thomas Spencer. *On the Decay of Correlations in $SO(n)$ -symmetric Ferromagnets*. *Communications in Mathematical Physics* **53** (1977), 299–302. <https://doi.org/10.1007/BF01609854>.
- [7] Robert Seiringer. *The Heisenberg ferromagnet: a dilute Bose gas in disguise*. Workshop contribution in *Mini-Workshop: New Directions in Correlated Quantum Systems*, Oberwolfach Report No. 7/2025 (2025), 357–358. <https://doi.org/10.4171/OWR/2025/7>.
- [8] Bálint Tóth. *Improved Lower Bound on the Thermodynamic Pressure of the Spin 1/2 Heisenberg Ferromagnet*. *Letters in Mathematical Physics* **28**(1) (1993), 75–84. <https://doi.org/10.1007/BF00739568>.
- [9] Daniel Ueltschi. *Random loop representations for quantum spin systems*. *Journal of Mathematical Physics* **54**(8) (2013), 083301. <https://doi.org/10.1063/1.4817865>.